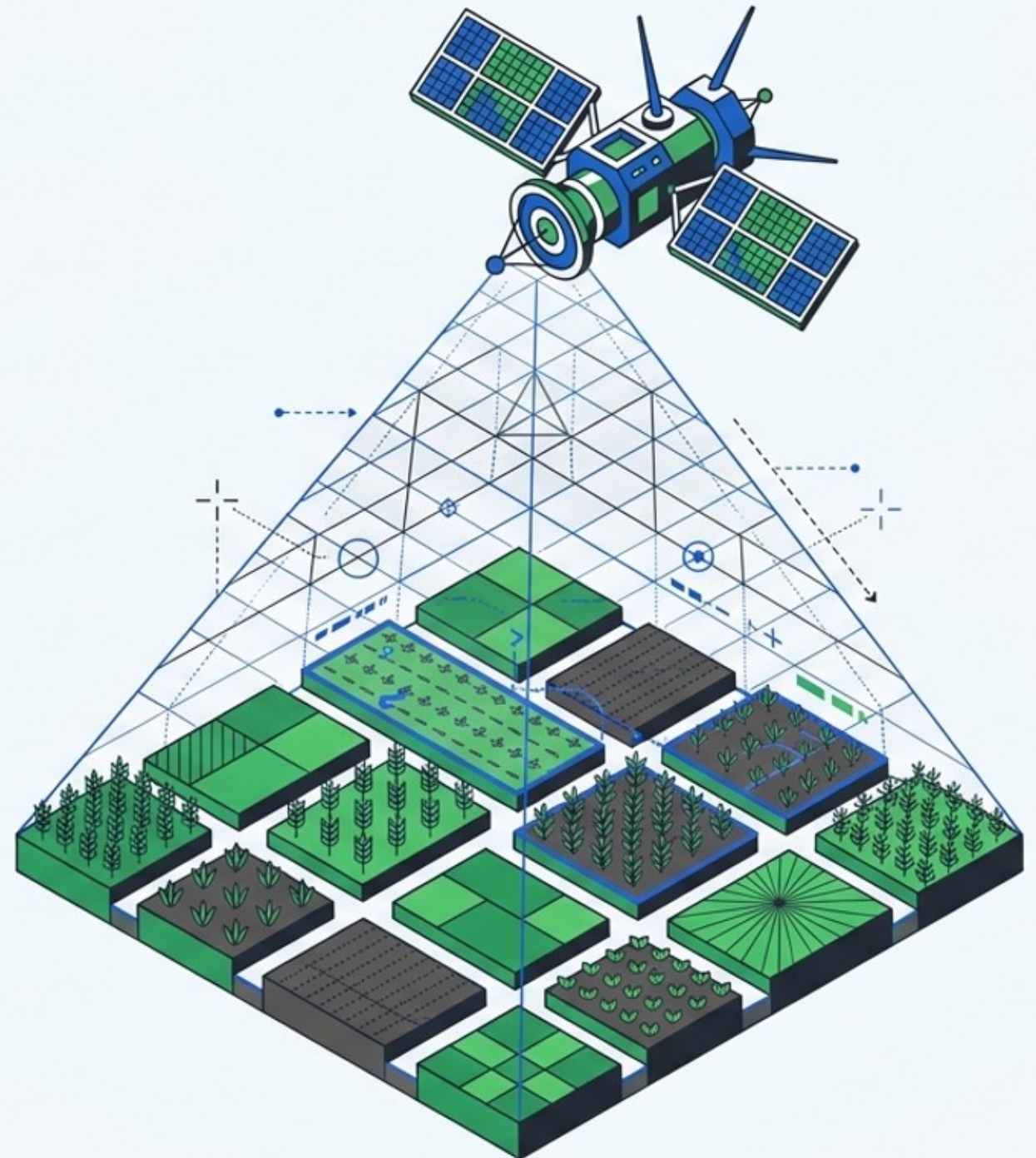




Multi-Sensor Satellite Data Fusion and Machine Learning-Based Crop Discrimination in Jhansi District, Uttar Pradesh, India

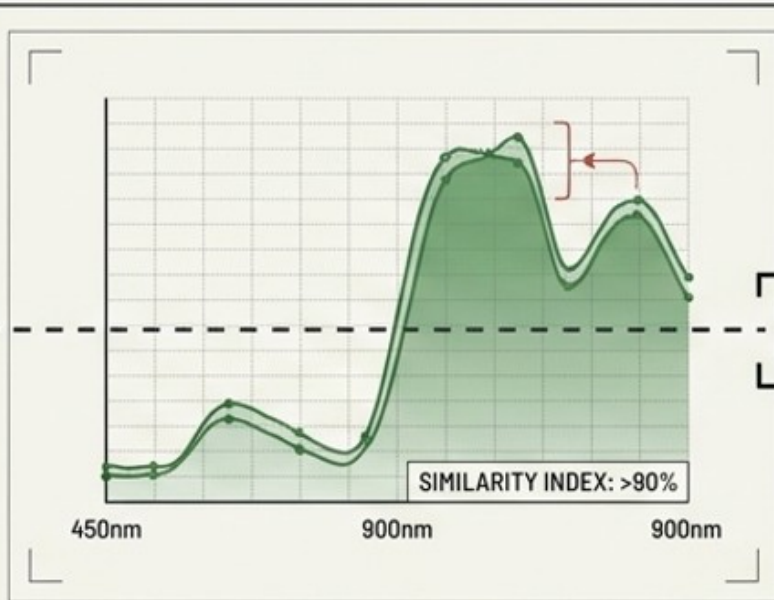
Submitted by: -
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Dhruv S Soni

Under the guidance of: -
Dr. Abhishek Dhanodia



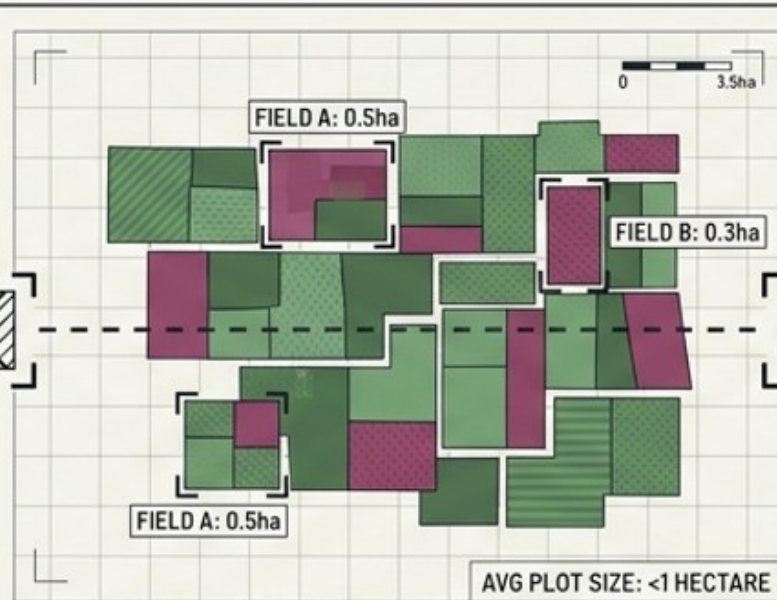
The Research Problem

Crop types are incredibly difficult to distinguish due to:



Obstacle 1: Spectral Confusion

Distinct crops like wheat and mustard exhibit highly similar spectral signatures in single-date optical imagery.



Obstacle 2: Spatial Fragmentation

Small field sizes and mixed cropping patterns common in Indian agriculture confound low-resolution sensors.



Obstacle 3: Operational Latency

Ground-based surveys are too slow for real-time food security intervention and policy planning.

Resolving this barriers requires multi – sensor fusion and continuous temporal analysis

Study Objectives

Compare Performance: Evaluate the capabilities of different satellite sensors.

Evaluate Modalities: Compare single-date versus temporal (time-series) data.

Compare ML Algorithms: Assess Random Forest, CART, GTB, and SVM.

Identify Best Model: Find the most robust model for crop classification.

Generate Accurate Maps: Produce reliable spatial crop distribution maps.

Area of Study



The Mission

Identify the most statistically robust combination of satellite platforms and machine learning algorithms to map fragmented winter crop cycles.



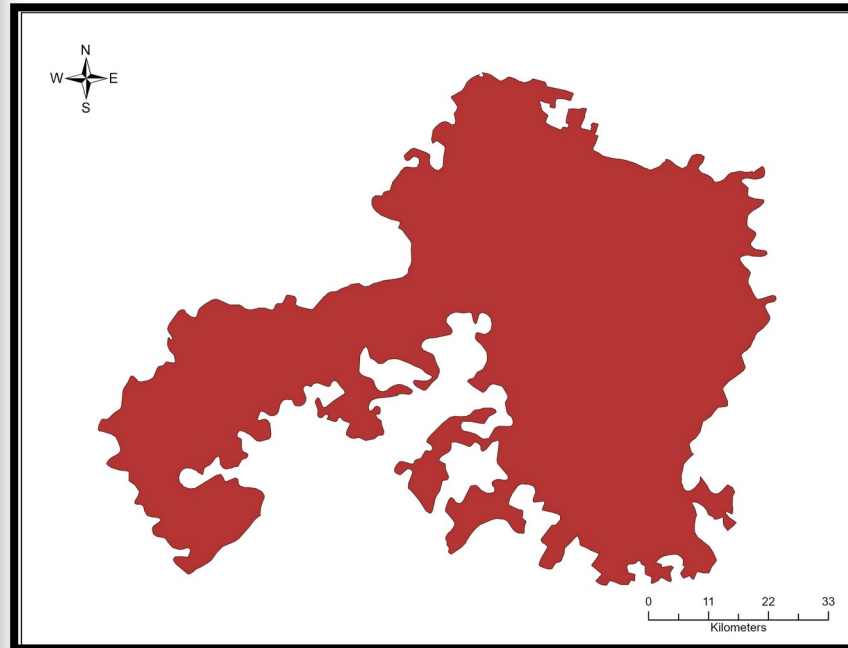
Crops Evaluated:

Wheat,
Mustard, Gram,
Field Pea



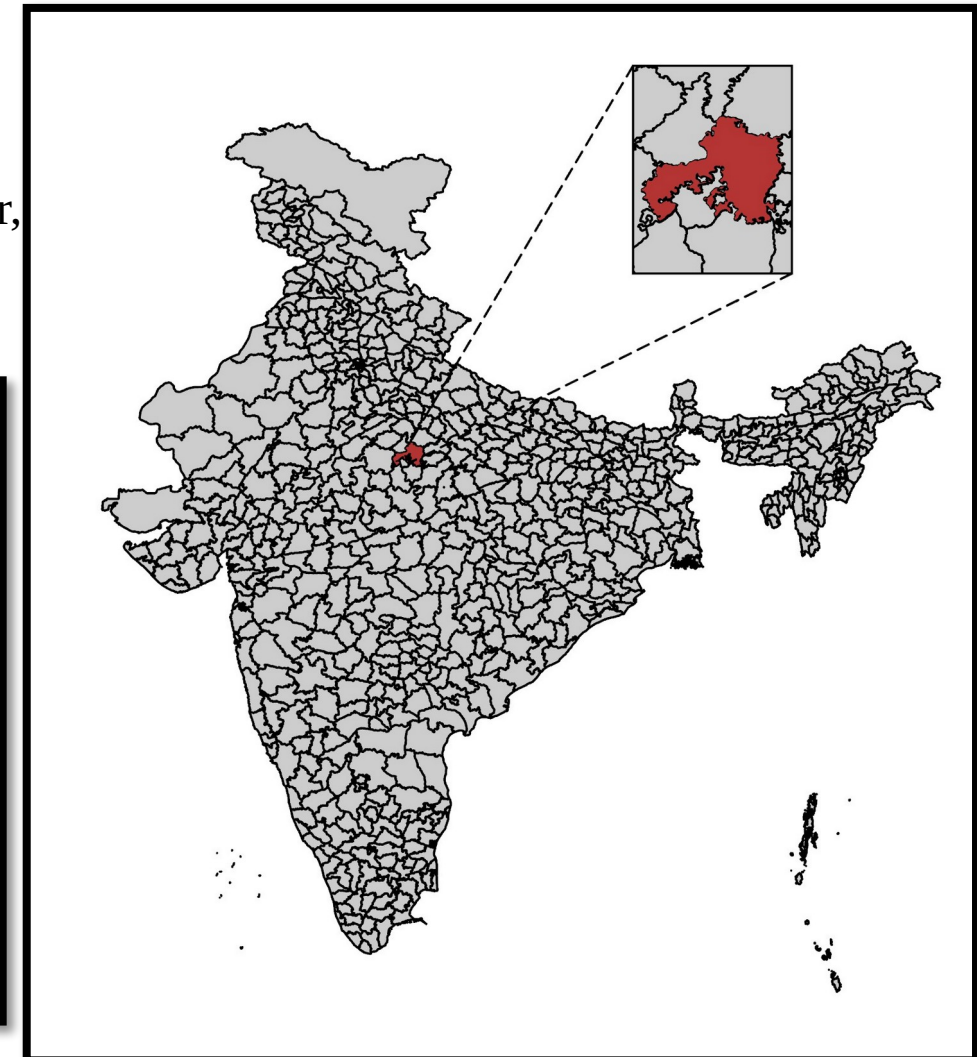
Other Classes:

Forest, Water,
Settlement,
Barren



Location: Jhansi District, Uttar Pradesh

Coordinates: ~25.45°N, 78.57°E



Satellites used



Sentinel – 1
(SAR)

Detects plant
structure and
moisture.
Completely cloud
- penetrating



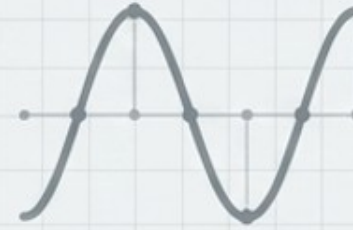
Sentinel – 2
(Optical)

Precision
chlorophyll
detection for high
– resolution
spectral
intelligence.



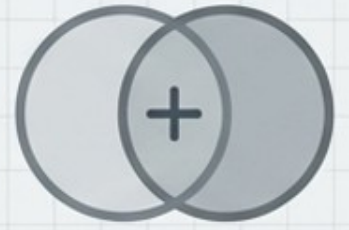
Landsat – 8
(Optical)

Long term multi –
spectral
consistency.



MODIS
(MCD43A4)

Coarse spatial
resolution, but
high temporal
frequency for
tracking
phenology.

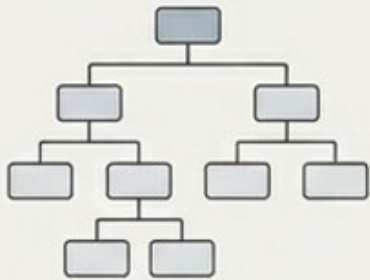


Data Fusion
(S1 + S2)

The ultimate
spectral –
structure hybrid
configuration.

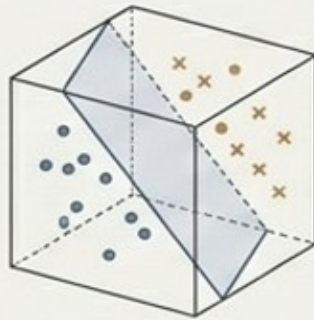
ML Algorithms used

CART



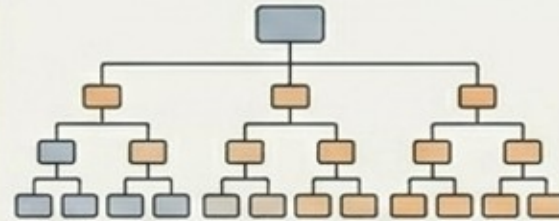
Simple operational logic.
Fast execution but highly
prone to overfitting.

SVM



Hyperplane division. Highly
effective for defining complex,
high-dimensional boundaries.

Random Forest (RF)



Parallel tree architecture.
An exceptionally robust and
stable ensemble method.

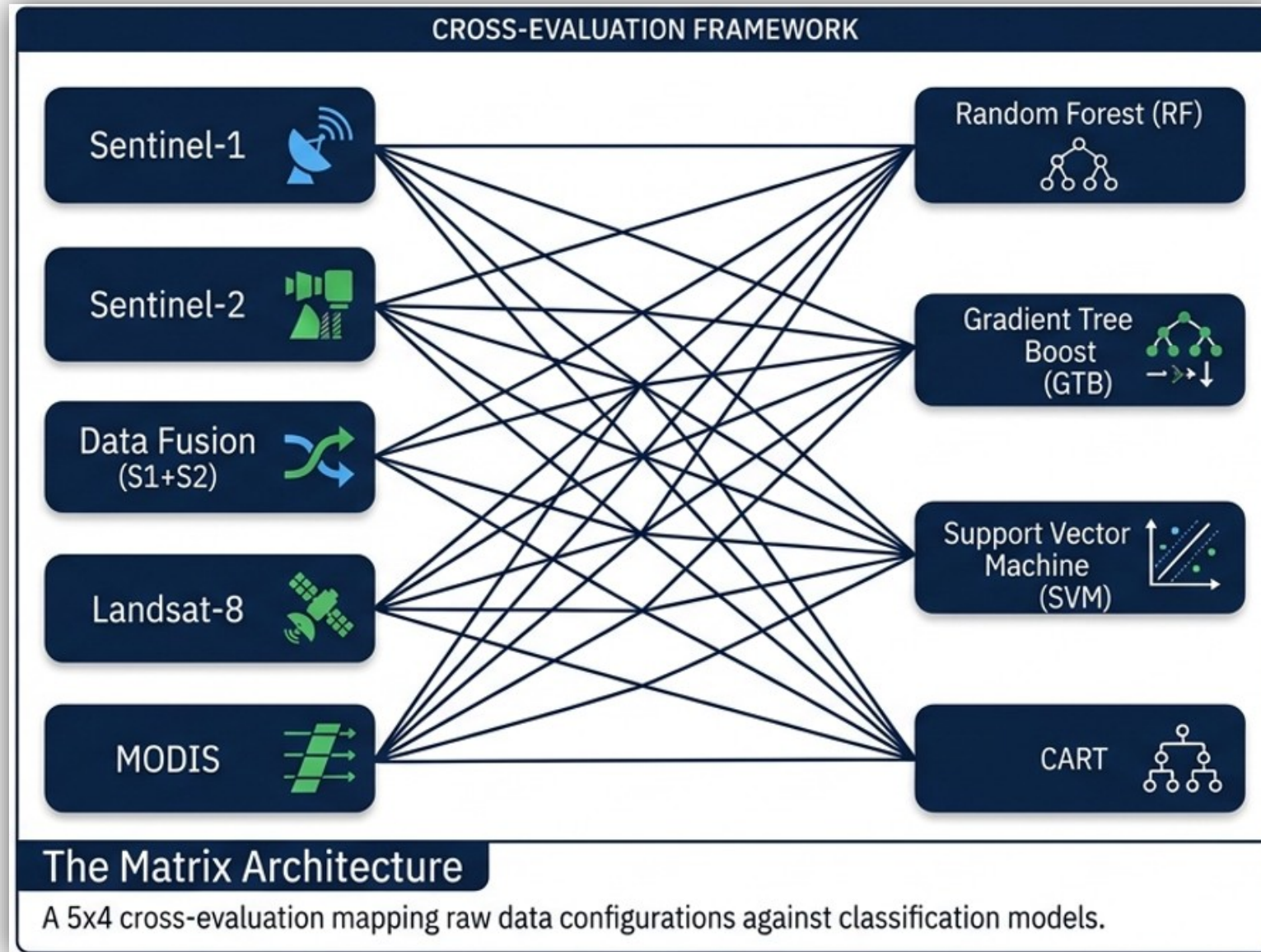
Gradient Tree Boosting (GTB)



Sequential error correction.
Yields the highest accuracy
through iterative optimization.

Ensemble methods performs better in this case

Main Idea



Features and Accuracy
Metrics used

Spectral Bands

Standard raw
reflectance
data across
multiple
wavelengths.

Vegetation Indices

**NDVI, EVI,
NDWI,
SAVI, LSWI**
used to track
health and
moisture.

SAR Features

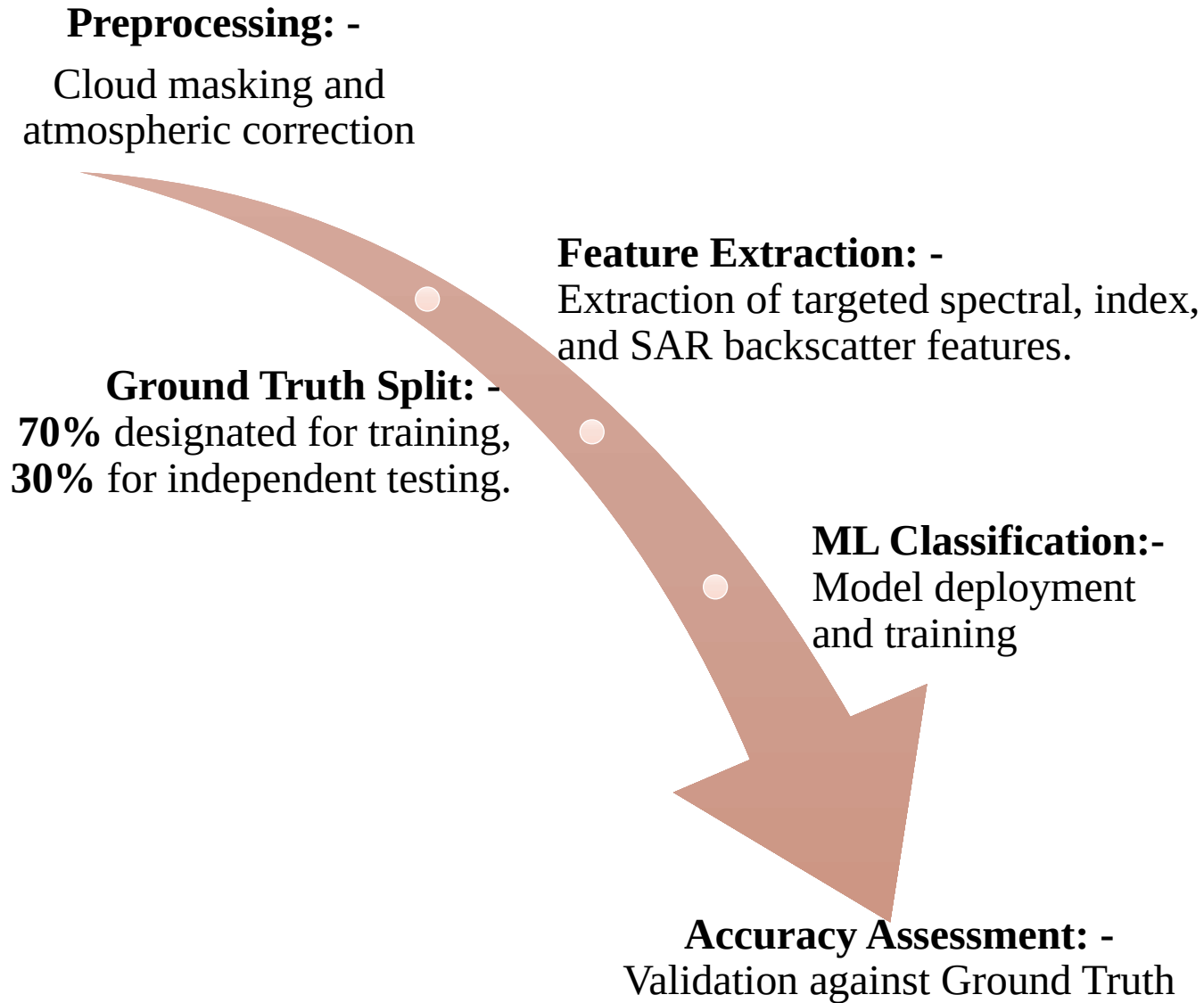
Structural
and
backscatter
metrics: **VV,
VH, and the
VV/VH
ratio.**

Accuracy Metrics

Overall
Accuracy

Kappa
Coefficient

Methodology





Important Findings

Temporal Performance



Temporal > Single date

Multi-date observations provide superior insights compared to single-day snapshots.

Data Synergy



Data fusion improves accuracy

Integrating Optical and SAR data sources leads to more robust and accurate classifications.

Model Selection



Tree-based models (RF, GTB)

Random Forest and Gradient Tree Boosting consistently outperform other algorithmic approaches.

Spatial Precision

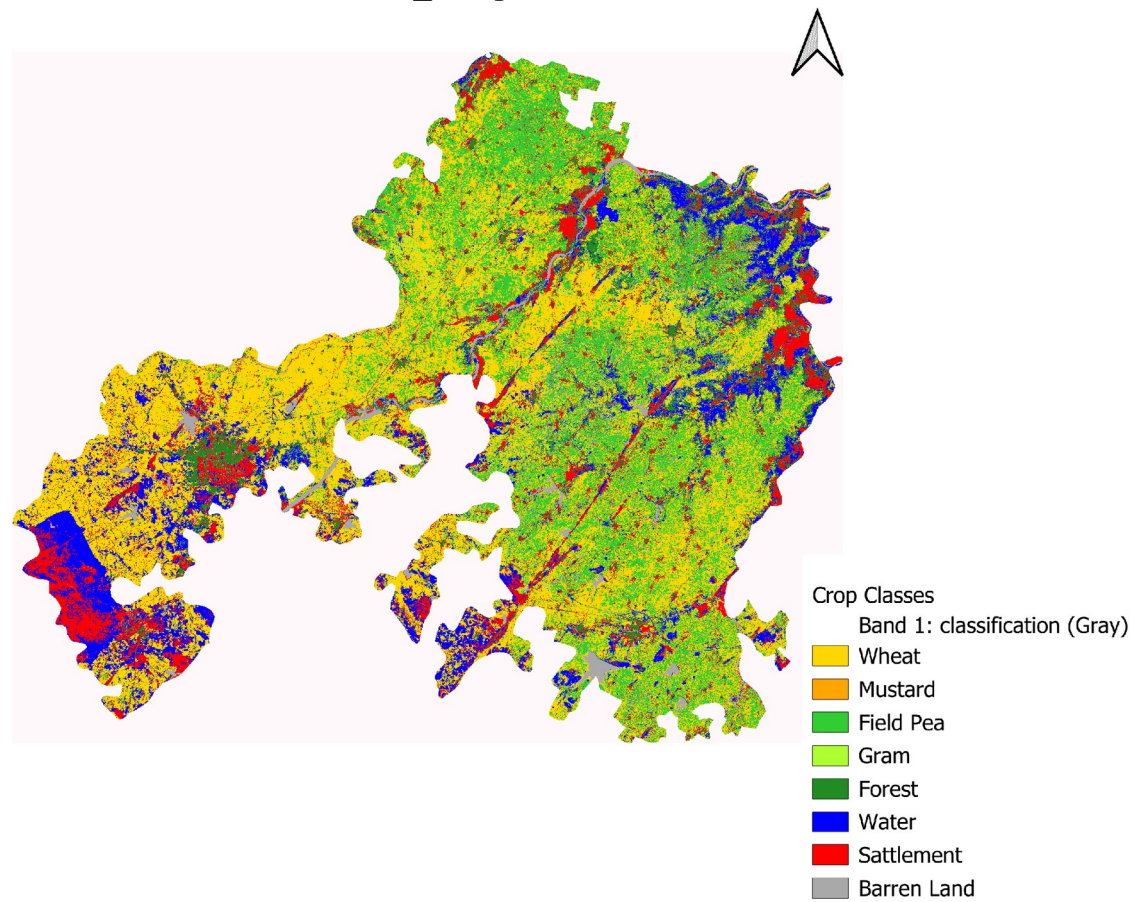


Spatial resolution is critical

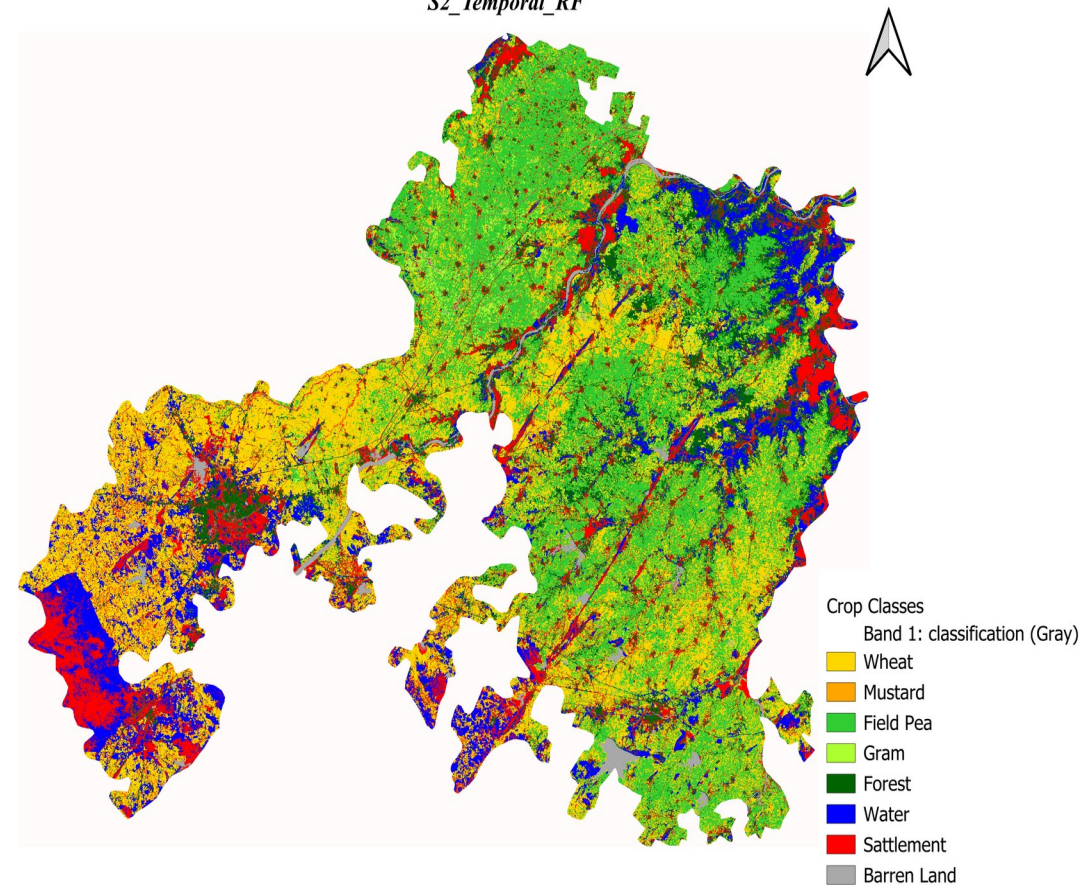
10m > 30m > 250m

Results

S1+S2_TemporalRF

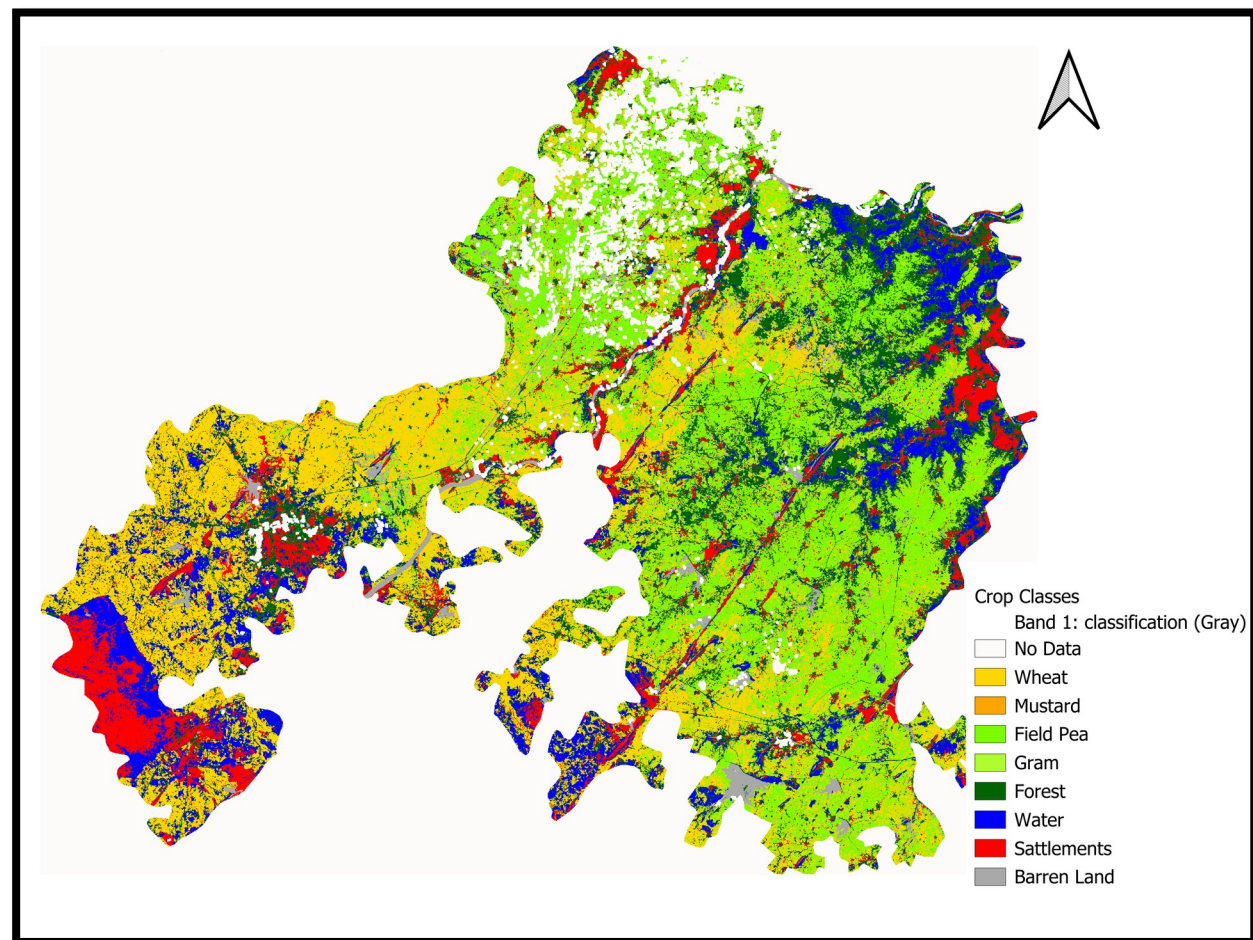
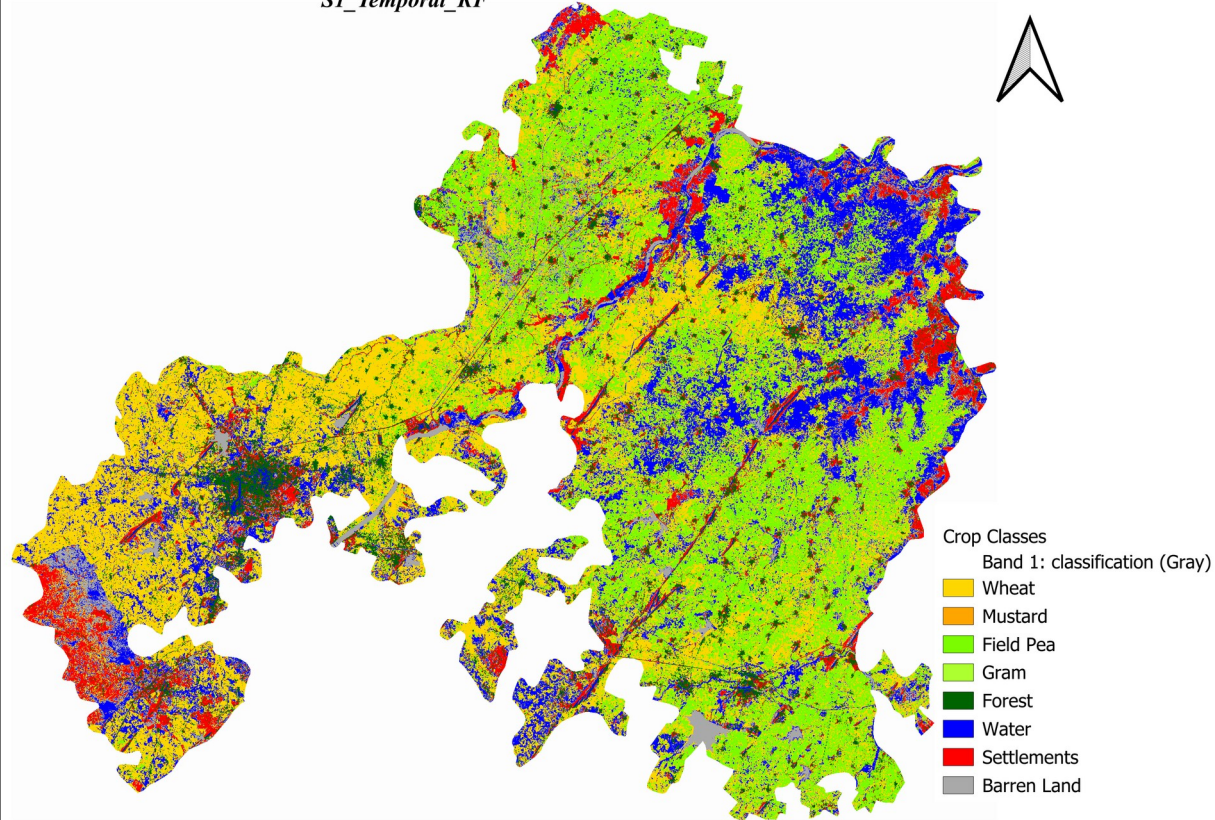


S2_Temporal_RF



Results

SI_Temporal_RF



Accuracy Matrix

Satellite / Sensor	Data Type	Algorithm	Overall Accuracy (%)	Kappa's Coefficient	Best Algorithm
Sentinel-1	Single Date	RF	57.56	48.54	SVM
		GTB	56.48	47.71	
		CART	50.75	40.86	
		SVM	62.81	53.52	
	Temporal	RF	77.52	72.46	RF
		GTB	73.92	68.52	
		CART	63.28	55.91	
		SVM	68.91	62.63	
Sentinel-2	Single Date	RF	81.07	76.87	GTB
		GTB	81.78	77.78	
		CART	78.21	73.35	
		SVM	80.00	75.44	
	Temporal	RF	85.00	81.70	RF
		GTB	83.92	80.52	
		CART	79.6	75.4	
		SVM	79.28	75.17	
MODIS	Single Date	RF	63.11	53.57	RF
		GTB	60.00	49.97	
		CART	54.66	43.83	
		SVM	48.88	35.52	
	Temporal	RF	59.45	50.06	GTB
		GTB	61.71	52.20	
		CART	54.50	44.62	

Accuracy Matrix

Satellite / Sensor	Data Type	Algorithm	Overall Accuracy (%)	Kappa's Coefficient	Best Algorithm
Landsat-8	Single Date	RF	78.92	74.43	RF
		GTB	73.92	68.75	
		CART	75.00	69.65	
		SVM	78.21	73.41	
	Temporal	RF	87.17	73.68	SVM
		GTB	84.61	68.42	
		CART	79.74	70.92	
		SVM	89.74	79.06	
Sentinel-1 + Sentinel-2	Single Date	RF	82.11	78.22	GTB
		GTB	83.21	79.54	
		CART	78.83	74.24	
		SVM	80.29	76.04	
	Temporal	RF	86.49	83.51	GTB
		GTB	87.22	84.43	
		CART	80.65	76.76	
		SVM	81.38	77.50	

Conclusion

Key Takeaways

- ✓ Multi-sensor + temporal approach is highly effective
- ✓ Sentinel-2 is best single sensor
- ✓ Fusion gives highest accuracy
- ✓ GTB is most powerful algorithm

Useful For:

-  Crop monitoring
-  Policy planning
-  Food security

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THANK YOU